Vissim lab: calibration and alternative designs

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Calibration (1/2): what was changed and why

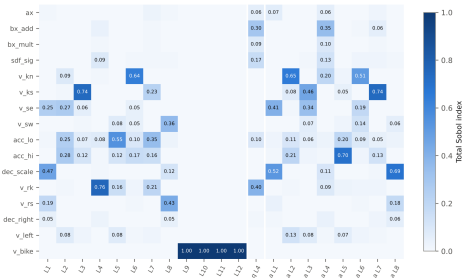
Approach: Python drives Vissim (COM); 1 857 runs in 3 phases

Latin hypercube → Gaussian-process surrogate per output → Sobol sensitivity → constrained optimum → validation with 10 new seeds

Most influential parameters

- right-turn speed (L4, L8, L1)
- desired speed per approach (exits L2, L3, L5, L6)
- desired acceleration / deceleration (L7, L1)
- bicycle desired speed (L9 to L12)
- conflict-area gaps: no influence

Constraints: demand served (GEH ≤ 5, L4 flow), 600 s warm-up added



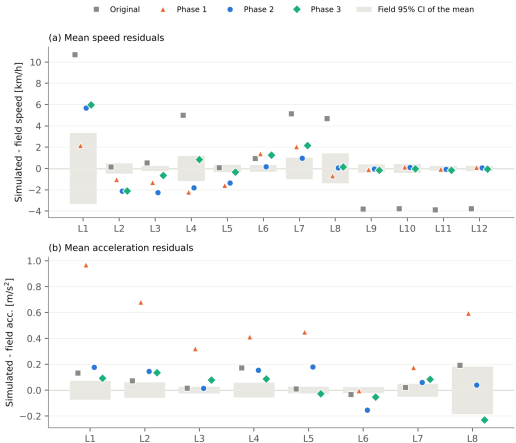
Total Sobol indices (surrogate, 567 configurations)

Calibration (2/2): outcome

	Original	Phase 1	Phase 2
Target		speed	speed + acc.
Speed RMSE [km/h]	4.53	**1.34**	2.01
Acc. RMSE [m/s^2]	**0.11**	0.53	0.13
Speed tests	2/12	6/12	**7/12**
Acc. tests	**7/8**	1/8	3/8
Max GEH	0.68	0.68	0.68

Calibrated (phase 2): right turns 10 km/h (was 22.5); bikes 17.3 km/h (was 13.5); Slottsgatan east 44 km/h; slightly tighter car following

Lesson: speed-only fitting (phase 1) used unrealistic braking; accelerations must be a target. 10 to 14 replications needed at L1/L8 ($\varepsilon = 5\,\%$).



Simulated minus field (10 new seeds); grey: field 95 % CI

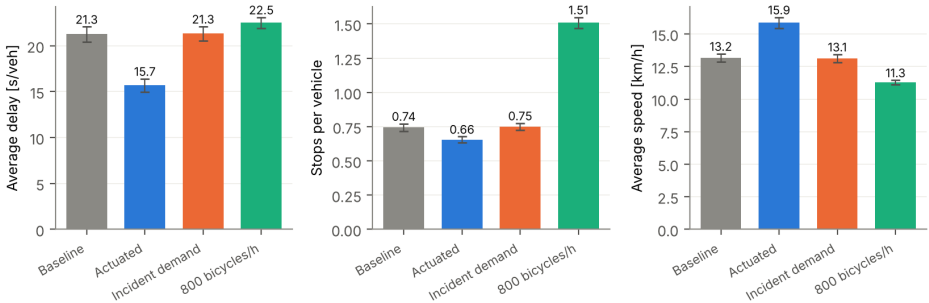
Alternative designs studied (all three)

Baseline = calibrated model (phase 2), fixed time program 1, constant demand. 20 replications per scenario with **common random numbers** → paired t-tests.

- **Actuated control**: VisVAP logic, gap 3 s on the detectors, stage 1 (Kungsgatan) up to 75 s, stage 2 (Slottsgatan) up to 15 s
- **Incident**: inputs A, C, E, G change every 10 min (Table 1); 30-min averages equal the baseline, so only the *timing* of demand changes
- **800 bicycles/h** per direction instead of 100

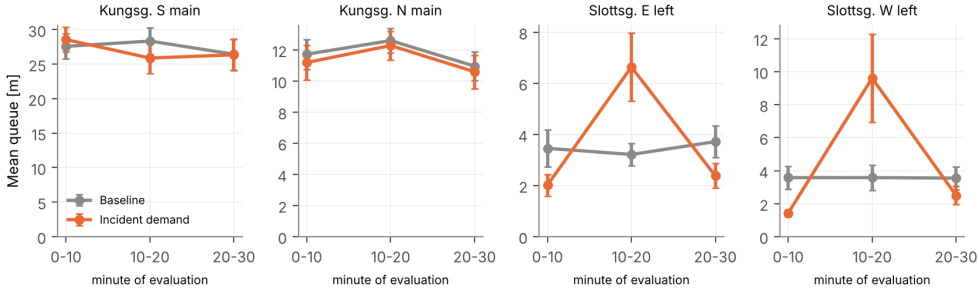
Metrics: average delay and stops per vehicle (level of service), queue length per approach in 10-min intervals (where and when), speeds and flows at L1 to L12 (link to the calibration, throughput)

Results (1/2): network performance and queues



	Actuated	Incident	800 bikes/h
Δ delay [s/veh]	-5.6 ($-26\,\%$, $p < 0.001$)	$+0.1$ ($p = 0.67$)	$+1.2$ ($p < 0.001$)
Queue Kungsg. northbound [m]	$27.5 \rightarrow 12.1$	$27.5 \rightarrow 26.9$	$27.5 \rightarrow 35.3$

Results (2/2): incident dynamics and bicycle conflict



Incident: Slottsgatan left-turn queues double to triple in minutes 10 to 20 and dissolve in the next interval; invisible in 30-min averages.

800 bikes/h: right-turners yield to cyclists; car flow at L4 943 $\rightarrow$ 878 veh/h ($-7\,\%$), stops doubled, bike speed 17.9 $\rightarrow$ 16.5 km/h.

Observations and questions for discussion

- The calibration target decides the result: speed-only fitting produced a model that was "right for the wrong reason". Should acceleration data be a standard calibration target?
- L1 has a field s.d. of 4.7 km/h but L3 only 0.3 km/h: an RMSE treats them equally, the t-test does not. Which acceptance criterion should a calibration use?
- Remaining misfits (L1 deceleration, exit speeds) look structural: turning speeds act only after the stop line. When should we change the model instead of its parameters?
- The actuated controller gains 26 % delay mainly by taking green from the side road. Is average delay the right objective, or should side-road waiting be weighted?
- Calibrated at 100 bicycles/h, but the bicycle scenario depends on car to cyclist conflicts that the data could not calibrate. How far can a calibrated model be extrapolated?